```
        pq.pop();

        Node* parent = new Node('$', left->freq + right->freq);
        parent->left = left;
        parent->right = right;

        pq.push(parent);
    }

    cout << "\nHuffman Codes:\n";
    printCodes(pq.top(), "");

    cout << "\nTime Complexity: O(n log n)";
    cout << "\nSpace Complexity: O(n)";

    return 0;
}
```

Output -

Output

```
Enter number of characters: 4
Enter character and frequency:
A 5
B 9
C 12
D 13

Huffman Codes:
A : 00
B : 01
C : 10
D : 11

Time Complexity: O(n log n)
Space Complexity: O(n)
```